

Biomask to heal injured face

UT Arlington engineers working with Army surgeons are developing a pliable polymer mask embedded with electrical, mechanical and biological components to speed up healing from disfiguring facial burns and help rebuild the faces of injured soldiers.

To help burn victims, Army physicians use polyethylene foam on the damaged tissue, which applies a vacuum to promote healing in the wounds. However, it could not be used on face because of the complex topography. The biomask is layered with sensors, actuators and medicine delivery tools. The patients are required to wear the mask for several months while their face regenerates.

The components allow localised monitoring and activation of treatment that can be applied to different parts of the wound as needed. The sensors provide feedback about the healing process, helping physicians direct appropriate therapy to different tissues.

The Biomask project is led by Eileen Moss—an electrical engineer and research scientist based at the UT Arlington Automation & Robotics Research Institute. Project partners include the US Army Institute of Surgical Research at the Brooke Army Medical Center in San Antonio and Northwestern University in Chicago.

Beware, computers can decode your thoughts!

University of California researchers have decoded electrical activity in the brain's temporal lobe—the seat of the auditory system—as a person listens to normal conversation. Based on the correlation between sound and brain activity, they were able to predict the words the person heard.

Researchers determined the location of intractable seizures in fifteen people undergoing brain surgery so



An X-ray CT scan of the head of one of the volunteers showing electrodes distributed over the brain's temporal lobe, where sounds are processed

that the area could be removed in a second surgery. Neurosurgeons cut a hole in the skull and placed electrodes on the brain surface or cortex—in this case, up to 256 electrodes covering the temporal lobe—to record the activity over a period of a week to pinpoint the seizures. The brain activity detected by the electrodes as patients heard five-ten minutes of conversation was recorded. The data was used to reconstruct and play back the sounds the patients heard.

Two different computational models were tested to match spoken sounds to the pattern of activity in the electrodes. The patients then heard a single word, and the models were used to predict the word based on electrode recordings.

Robotic arm controlled with monkey's thoughts

As a monkey manipulates objects with his hands, not far away a robot hand mirrors his fingers' moves as it receives instructions from the chips implanted in his brain.

Zheng Xiaoxiang of the Brain-Computer Interface Research Team

at Zhejiang University in Zijingang, China, and colleagues have successfully captured and deciphered the signals from the monkey's brain and interpreted them into the real-time robotic finger movements.

The two sensors implanted in the monkey's brain monitor just 200 neurons in his motor cortex. However, this was enough to accurately interpret the monkey's movements and control the robotic hand.

Humans have used electrodes to control prosthetic arms, but Zheng claims this research looks at the finer movements of the fingers.

"Hand moves are associated with at least several hundreds of thousands of neurons. We now decipher the moves based on the signals of about 200 neurons. Of course, the orders we produced are still distant from the truly flexible finger moves in complexity and fineness," she said.

Paper robots

Paper structures built using the principles of origami could lead to cheap, easy-to-make robots that could be used to provide extra hands for surgeons or handle delicate objects such as eggs or fruit.

George Whitesides and colleagues at Harvard University had previously built squid-inspired robots with artificial muscles made from soft plastic and powered by pneumatic air pumps. Now they have combined this technique with paper to create a series of lightweight structures capable of bending and twisting, and even lifting heavy weights.

Paper is flexible, but unlike plastic it does not stretch, making it useful for forming rigid structures when a paper balloon is filled with air. These simple designs could be improved upon to create 'soft' robots able to work closely with humans, unlike some robots currently used on factory assembly lines.